



RUDRAEDGE AEROSPACE

Project - Rudra Edge RE-T1(Spectra) Daily Engineering Log

1. TEAM MEMBER INFORMATION

Engineer Name: Suraj Nadda

Designation / Role: Software & Data Engineer

Task Order Reference: 9 October 2026 – Data Validation & Error check

2. DAILY LOG ENTRY FORM

- **Date:** 09 October 2026

- **1. Assigned Task:**

Data validation, boundary limits, and corrupted packet detection.

- **2. What Was Studied:**

Studied the purpose of validating telemetry data before it is used in further processing. Reviewed common data-quality problems, including missing sensor readings, incorrect numeric values, out-of-range measurements, invalid timestamps, duplicate packet IDs, and records arriving out of chronological order. Also studied how validation rules and rejection reasons help identify unreliable records and support traceable offline data processing.

- **3. What Was Implemented:**

Created and ran a Python validation script using Pandas to inspect a sample telemetry dataset containing timestamp, packet ID, temperature, pressure, altitude, and speed fields. Implemented checks for required columns, missing or non-numeric sensor values, configured demonstration boundaries, invalid timestamps, duplicate packet IDs, missing identifiers, and out-of-order timestamps. Added error reasons and separated accepted records from rejected records, then generated CSV outputs and a validation summary.

- **4. Tools / Libraries Used:**

VS Code, PowerShell terminal, Python 3.14.6, Pandas 3.0.6, CSV files

- **5. Output File Name-** suraj_rudra_task/Day5/codes/telemetry_validation.py
Path-https://github.com/surajnadda29/suraj_rudra_t1.git

- **6. Problems Encountered:**

No significant problems were encountered during the implementation.

- **7. How Problem Was Resolved:**

. No significant implementation problem encountered

- **8. Next Task Scheduled:**

(10 October 2026) – Telemetry Schema Design (timestamps, sensors IDs, status mask)

- **9. Personal Expenditure: ₹0 (Verified Mandate)**

3. ENGINEER SELF-DECLARATION & SIGN-OFF

I hereby formally declare that today's technical progression, synthetic dataset code, and local repository paths have been documented truthfully in compliance with Section 6 (Common Technical Standards). I certify with absolute transparency that my personal financial expenditure for today's assigned task is exactly ₹0, utilizing strictly local open-source infrastructure.

Candidate/Engineer Signature:  _____

Date: 09/10/2026

Full Name: Suraj Nadda

PROJECT LEAD VERIFICATION BOX — TO BE FILLED BY SARVESH SHARMA ONLY

I have reviewed today's technical output, local repository commits, and verified the project compliance standards for this day's assigned task.

Task Evaluation Status:

- ☐ Approved / Completed
- ☐ Needs Revision / Delayed
- ☐ Technical Block Encountered

Operational Expenditure Confirmed: ₹0 (Strict Zero-Cost Compliant)

Remarks/Notes: _____

Project Lead Signature: _____ Date: ____/____/2026